

# ASSESSMENT TOOL 2

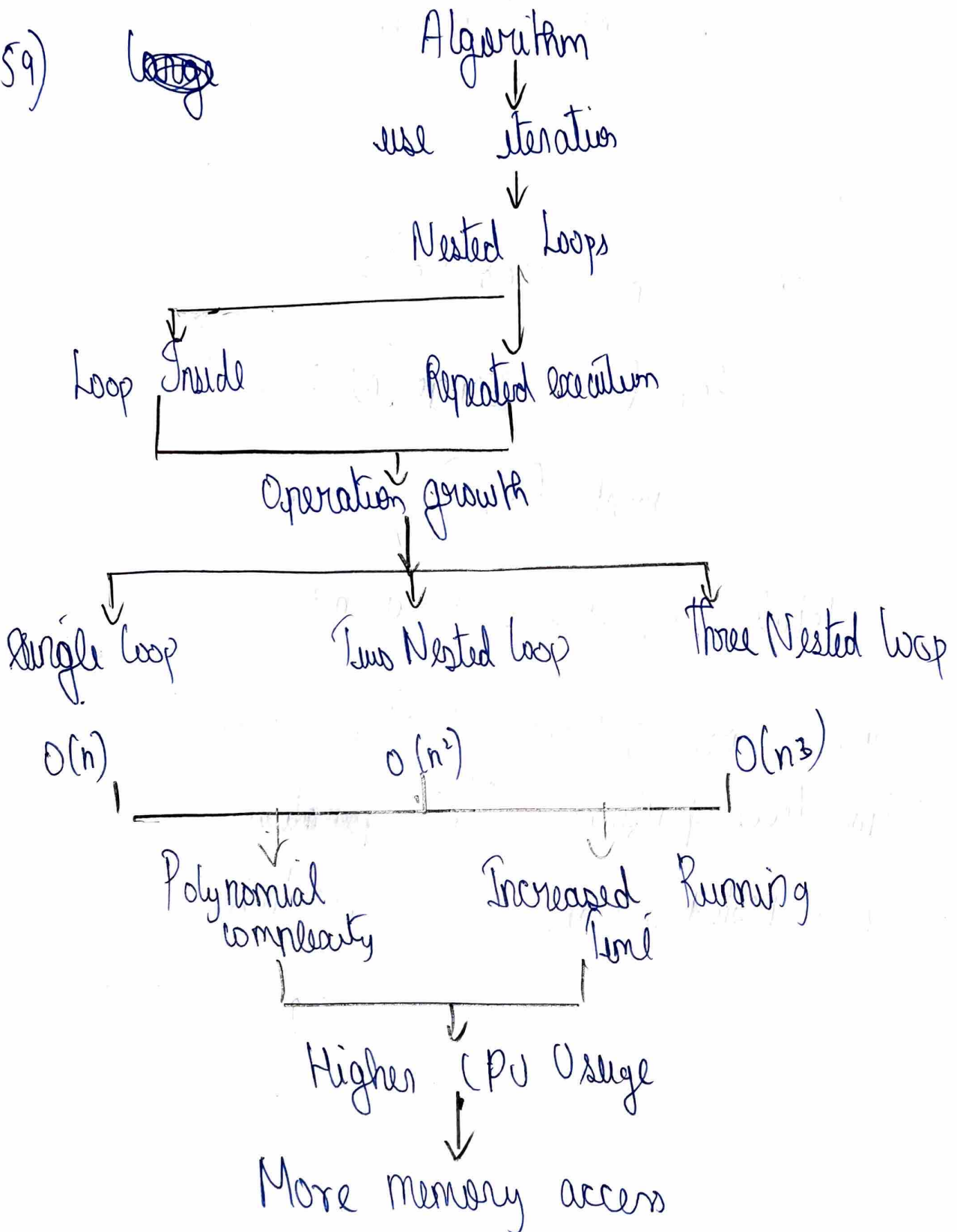
## CONCEPT MAPPING

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59)

~~Concept~~



Poor Scalability

↓  
Reduced performance

↓  
Inefficient Algorithm design

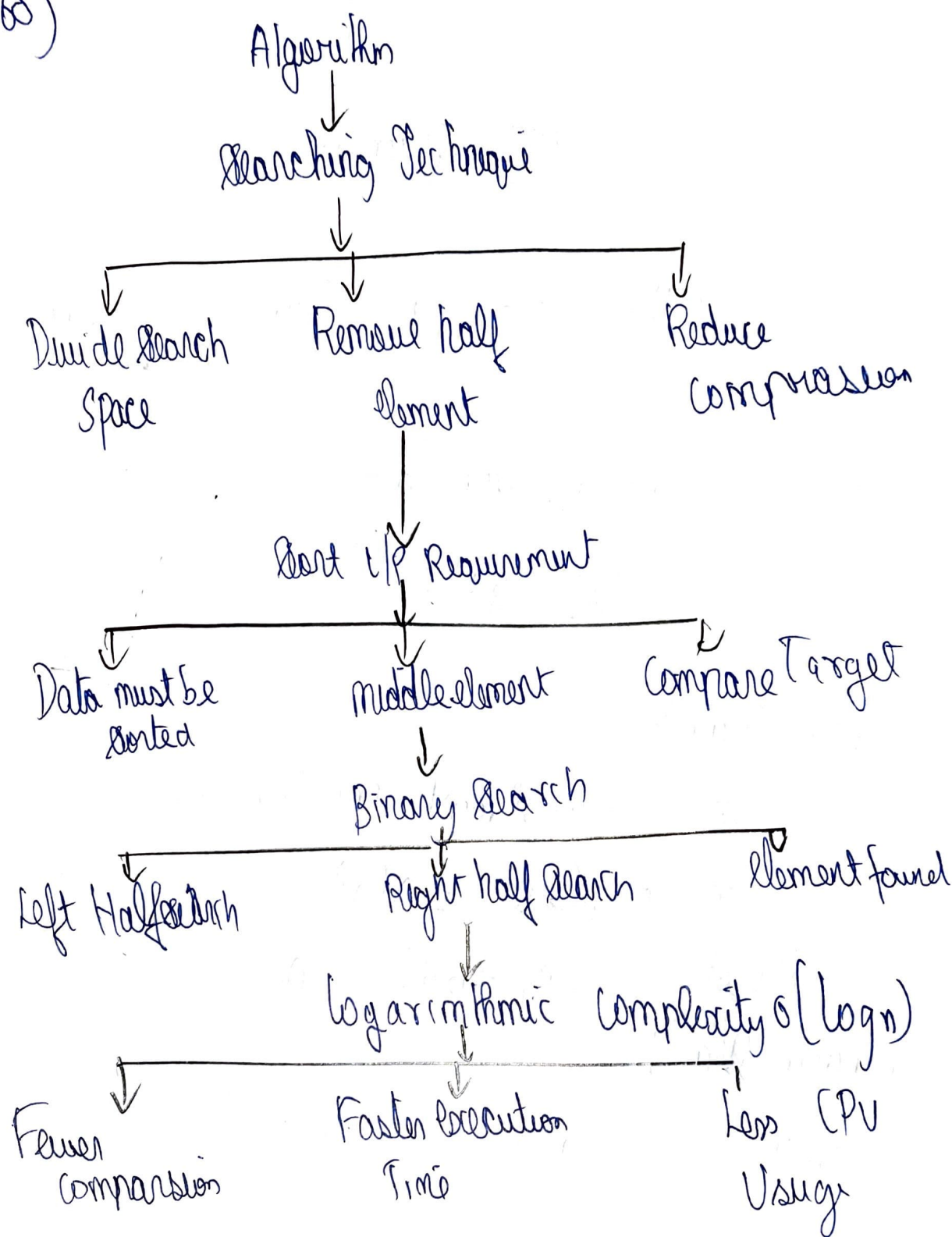
Python

```
for i in range(n):  
    for j in range(n):  
        print(i, j)
```

Total Operation =  $n \times n = n^2$

- \* Single loop perform  $O(n)$  operation
- \* Two loop performs  $O(n^2)$  operation
- \* Three nested performs  $O(n^3)$  operation

60)



Better Scalability



High efficiency



better performance than linear search



suitable for large sorted

Data set

Linear Search

- \* Checks every element one by one

- \* Time complexity =  $O(n)$

- \* Worst case comparison =  $n$

Binary Search

- \* Eliminates half element in every step

- \* Time complexity =  $O(\log n)$

- \* Worst Comparison =  $\log_2 M$